Northern University of Business and Technology, Khulna
Department of Computer Science and Engineering
CSE 4104 — Software Development III

PROJECT PROPOSAL

AI Academic Mentor — Personalized Learning & Study Assistant

Team Name	CSE4104-7D-T06
Section	7D
Project Title	AI Academic Mentor — Personalized Learning & Study Assistant
Course Code	CSE 4104
Submission Week	Week 2 — Project Proposal
Date	9 November 2026
Prepared By	Team CSE4104-7D-T06
Supervised By	Md. Riaz Mahmud, Assistant Professor, CSE Dept., NUBTK

1. Project Title

AI Academic Mentor — Personalized Learning & Study Assistant

2. Problem Statement

Students in university often face the following challenges:

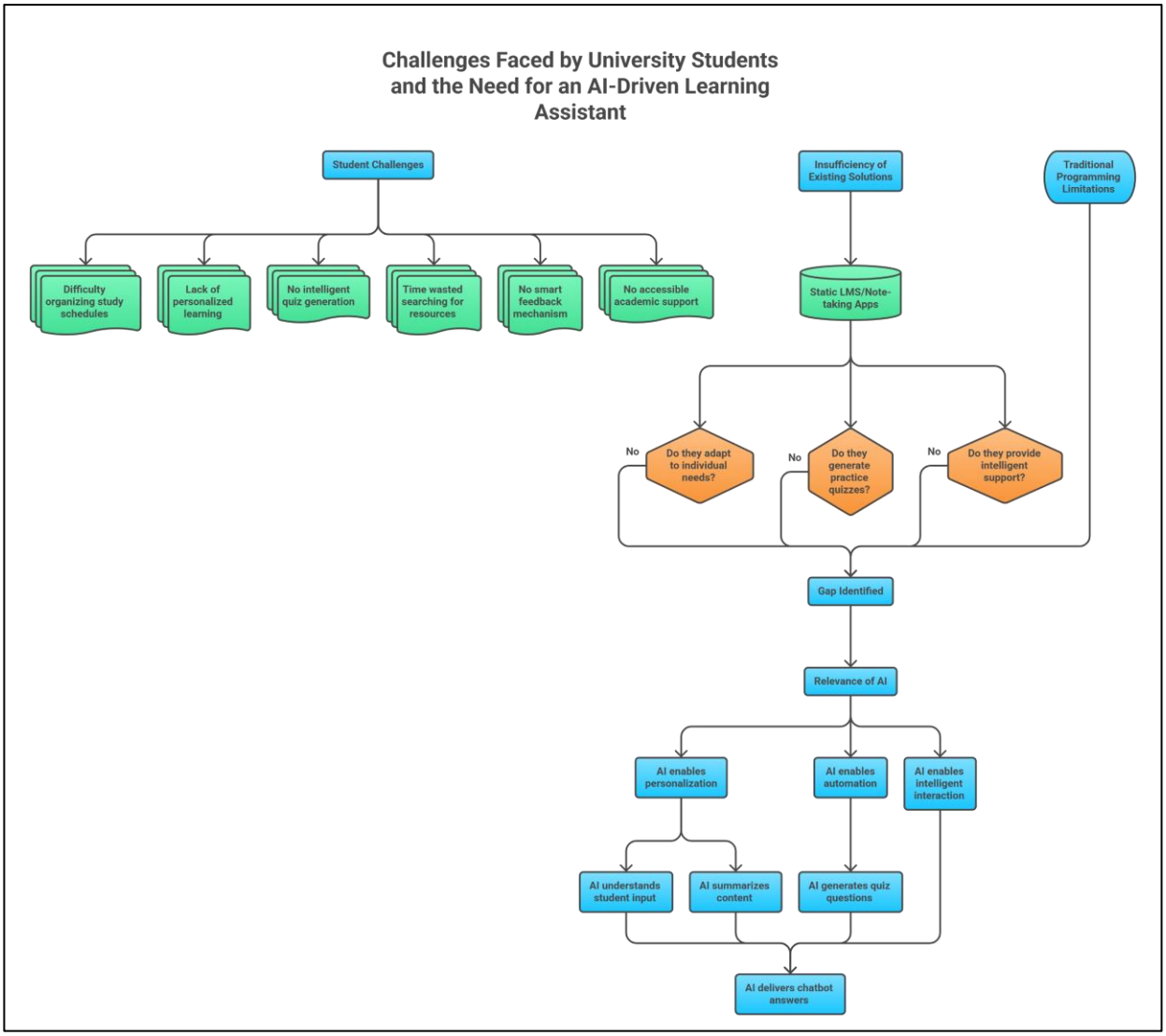
- Difficulty organizing and managing their study schedules efficiently across multiple subjects.
- Lack of personalized learning — every student receives identical content regardless of their individual weak areas or learning pace.
- No intelligent system to generate quizzes from study material or specific topics on demand.
- Significant time wasted searching for learning resources instead of focusing on active, effective studying.
- No smart feedback mechanism to track academic progress and highlight areas requiring improvement.
- No affordable, accessible 24/7 academic support outside of scheduled class hours.

Why existing solutions are insufficient:

Existing systems such as simple LMS platforms (e.g., Google Classroom) or generic note-taking apps are entirely static. They store and deliver content but do not adapt to individual student needs, do not generate practice quizzes automatically, and do not provide any intelligent conversational academic support. There is a clear and pressing gap for an intelligent, AI-driven learning assistant that personalizes the study experience, auto-generates quizzes, provides instant academic chatbot support, and summarizes notes — all within a single integrated platform.

Why AI is relevant for solving this problem:

AI is the foundational layer that makes personalization, automation, and intelligent interaction possible. Without AI, the system would be a static CRUD application with no adaptive capability. AI enables the system to understand student input, generate relevant quiz questions, produce concise summaries of any content, and deliver contextual answers through a chatbot — none of which is achievable through traditional programming alone.



3. Objectives

The primary objectives of AI Academic Mentor are:

- To build a full-stack AI-integrated web application that serves as a personalized study assistant for university students.
- To allow students to register, log in, and maintain a personalized study dashboard tracking their activity and progress.
- To integrate an AI chatbot (using Gemini API or OpenAI API) that answers academic questions intelligently and contextually.
- To automatically generate multiple-choice and short-answer quizzes from any given topic or subject.
- To summarize study notes or uploaded content using AI summarization to accelerate student revision.
- To generate a personalized daily study plan based on each student's subjects, goals, and available study time.
- To track student progress visually and highlight weak areas through a dashboard with charts and analytics.
- To improve student productivity by reducing time spent searching for resources and managing study schedules manually.
- To deploy the application online as a production-ready, publicly accessible web system with proper security.

4. Proposed Features

4.1 Authentication System

- User Registration and Login with form validation
- JWT-based secure session management
- Password hashing with bcrypt
- Role-based access: Student and Admin

4.2 AI Chatbot

- Students can ask any academic or subject-related question
- AI generates contextual, accurate, and beginner-friendly answers
- Full conversation history stored per student session
- Powered by Gemini API or OpenAI GPT

4.3 AI Quiz Generator

- Enter any topic and instantly receive an AI-generated quiz
- Supports both MCQ and short-answer formats
- Quiz scores tracked and stored in database for progress analysis

4.4 AI Note Summarizer

- Paste or upload study notes directly into the platform
- AI generates a concise, structured, easy-to-memorize summary
- Helps students revise faster and retain information more effectively

4.5 Personalized Study Planner

- Students enter their subjects and weekly available study hours
- AI generates a structured, day-by-day personalized study plan
- Plans saved to the database and updatable at any time

4.6 Progress Dashboard

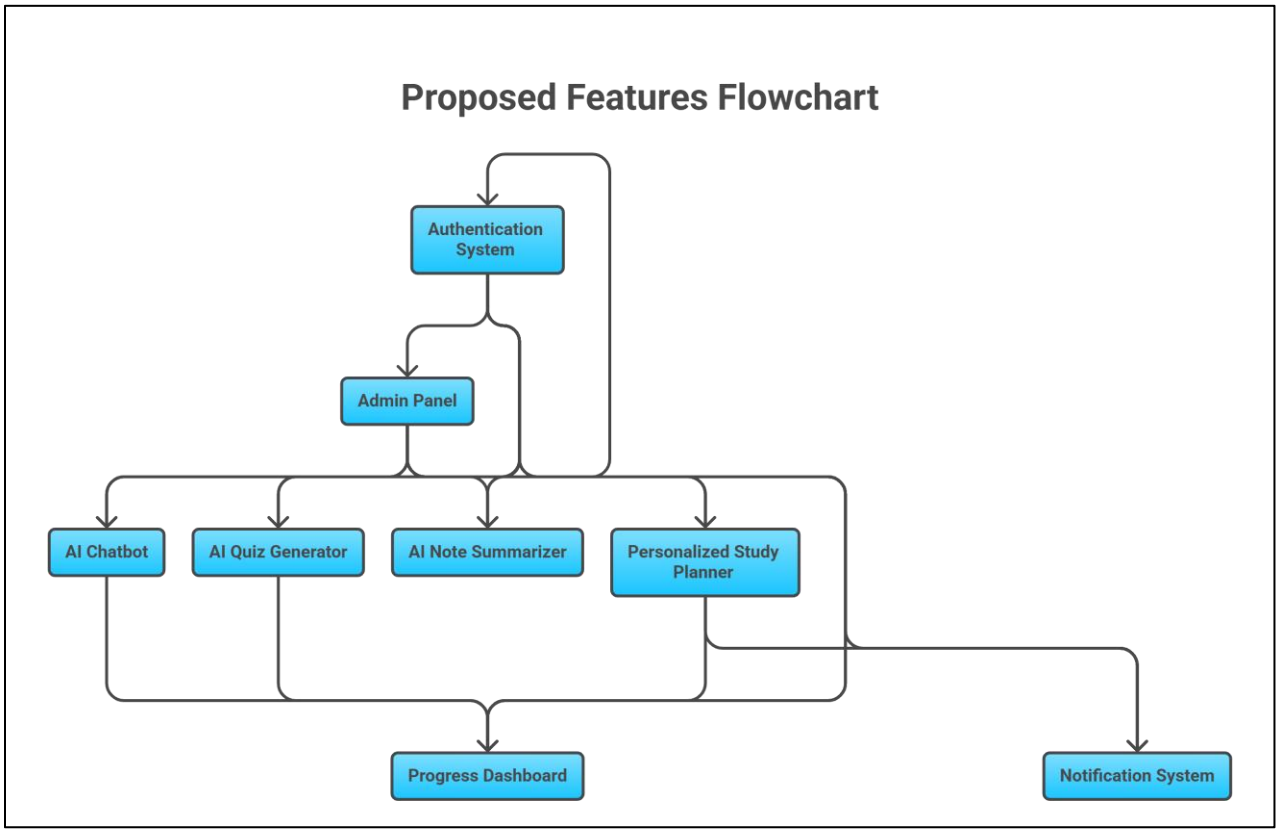
- Visual charts showing quiz scores and performance trends over time
- Topic-wise performance tracking to identify weak areas
- Weekly study activity summary with engagement metrics

4.7 Notification System

- In-app notifications for study plan reminders
- Alerts for pending quizzes and upcoming review sessions

4.8 Admin Panel

- View all registered users and their activity data
- Monitor overall system usage and platform health
- Manage content and post announcements to all students



5. Technology Stack

Layer	Technology	Purpose
Frontend	React.js + Tailwind CSS	User interface and responsive design
Backend	Node.js + Express.js	REST API, business logic, authentication
Database	MongoDB + MongoDB Atlas	Flexible data storage and cloud hosting
AI Integration	Gemini API / OpenAI API	Chatbot, quiz generation, summarization
Authentication	JWT + bcrypt	Secure login and session management
Frontend Deploy	Vercel	Fast, free frontend hosting
Backend Deploy	Render / Railway	Free backend API hosting
Version Control	GitHub	Code management and team collaboration
UI Design	Figma	UI/UX wireframes and prototyping
API Testing	Postman	Testing and documenting REST APIs

6. Expected Outcome

What users will gain from the system:

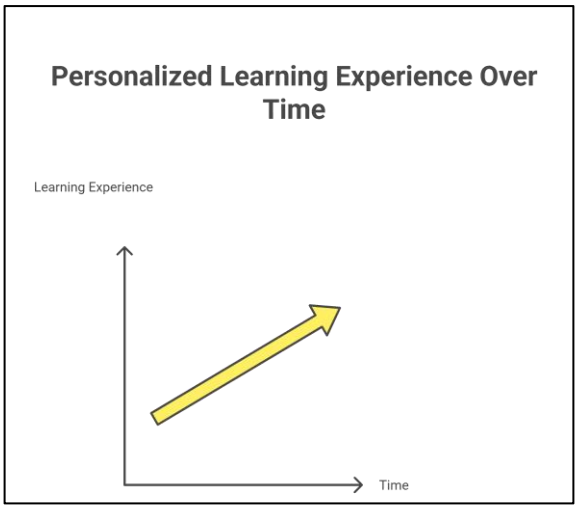
- A fully functional, deployed AI-powered web application accessible from any browser at any time.
- A genuinely personalized learning experience that adapts to each student's individual needs, weak areas, and study goals.
- Instant 24/7 academic support through the AI chatbot, eliminating dependency on instructor availability outside class.
- Automated quiz practice on any topic, saving time and making self-assessment effortless.
- Faster, more efficient note revision through AI-generated summaries of dense study material.

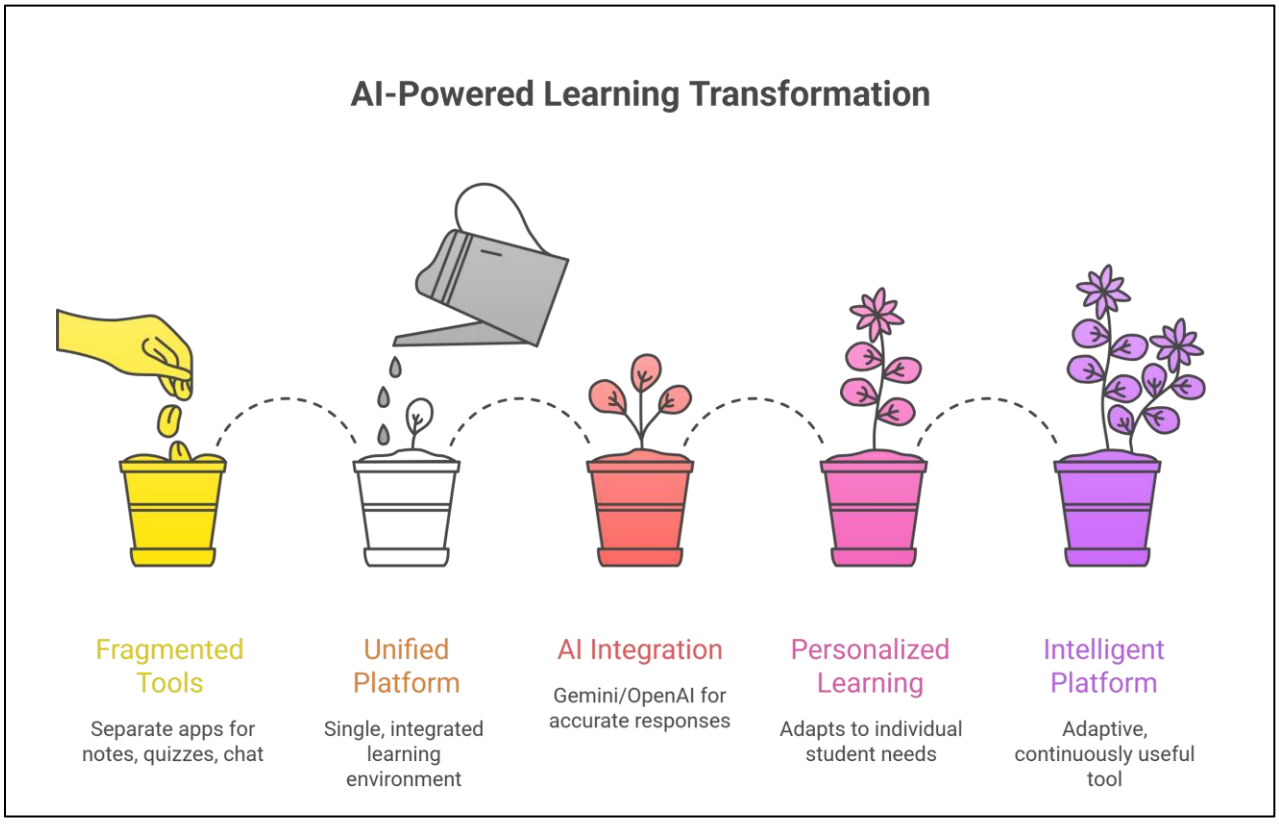
How the system improves existing workflow:

- Replaces fragmented tools (separate apps for notes, quizzes, chat, planning) with a single unified platform.
- Eliminates the manual effort of creating study schedules through AI-powered automated planning.
- Provides data-driven insight into student performance through visual progress tracking — not available in current LMS tools.

Expected benefits of AI integration:

- AI transforms a static academic platform into an intelligent, adaptive, and continuously useful tool.
- Meaningful prompt engineering ensures responses are accurate, contextual, and appropriate for university-level students.
- Real API integration (Gemini/OpenAI) ensures production-quality AI output, not simulated responses.





7. Initial Project Planning

Week	Phase	Deliverable
Week 1	Team Formation & Idea Selection	Team info, GitHub repo, technology selection
Week 2	Project Proposal	Proposal PDF (this document)
Week 3	SRS Preparation	SRS document + use case diagrams
Week 4	System Design	Architecture diagram, ER diagram, API structure
Week 5	UI/UX Design	Figma wireframes for all major pages
Week 6	Backend Development	Authentication system, REST APIs, database setup
Week 7	Frontend Development	All UI pages built and connected to backend APIs
Week 8	AI Integration	Chatbot, quiz generator, note summarizer integrated
Week 9	Feature Completion	All features working, UI polished and refined
Week 10	Testing & Debugging	Testing report, bug fix documentation
Week 11	Deployment	Live deployment on Vercel + Render / Railway
Week 12	Documentation	Final report, user manual, API documentation
Week 13	Presentation Prep	Slides prepared and demo rehearsed
Week 14	Final Presentation & Viva	Live demonstration and Q&A evaluation

8. GitHub Repository

GitHub Repository Instructions followed: Repository created, all team members added as collaborators, initial project structure uploaded, proper README.md included.

Repository Name	cse4104-7c-t05-ai-academic-mentor
Repository URL	https://github.com/Santu-Mondal/AI-Academic-Mentor-Personalized-Learning-Study-Assistant/tree/411142fcd5c145a92f334e7a9af446e864ef6c69/documentation
Branch Structure	main / development / frontend / backend / ai-integration

README.md includes:

- Project title: AI Academic Mentor — Personalized Learning & Study Assistant
- Team information: all member names, IDs, and roles
- Proposed features: authentication, AI chatbot, quiz generator, note summarizer, study planner, progress dashboard
- Technology stack: React.js, Node.js, MongoDB, Gemini/OpenAI API, Vercel, Render / Railway
- Project description: problem statement, objectives, and expected outcome
- Installation instructions and deployment link (to be updated upon deployment)

Evaluation Criteria Addressed

Problem selection quality	Real, locally relevant problem facing university students in Bangladesh — not a generic or trivial challenge.
Innovation	Three AI features in one integrated platform; personalized study planner; progress analytics — no comparable local system exists.
AI relevance	AI is core to the system, not decorative. Chatbot, quiz generator, and summarizer all require real AI API integration with meaningful prompt engineering.
Proposal clarity	All six required sections clearly documented: title, problem statement, objectives, features, technology stack, expected outcome.
Feasibility	Full-stack system using well-documented, freely available tools (React, Node.js, MongoDB, Gemini/OpenAI) within a 14-week timeline.
Project planning quality	Complete 14-week development timeline with specific deliverables per phase aligned to the course handbook.

Declaration

We, the undersigned team members of CSE4104-7D-T06, hereby declare that this project proposal is our original work. We understand the academic integrity policy of CSE 4104 and commit to following all course guidelines throughout the semester. AI integration in this project will use real APIs with meaningful prompt engineering — not simulated or fake AI output.

Role	Name	Student ID	Signature
Team Leader	Efaz Rahman	11230121162	
Frontend Developer	Santu Mondal	11230321312	
Backend Developer AI Integration Lead	MD Ruhan Tanveer	11230321335	

Date: 09 November 2026